Efe Onaran

Contact Information

University of Pennsylvania
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Research Interests

Probability theory with a focus on the topology and geometry of random high-dimensional objects, and random growth models.

Academic Appointments

University of Pennsylvania 2024–present
- Postdoctoral Researcher 2024–present
  Department of Electrical and Systems Engineering
- Visiting Scholar, Department of Mathematics 2024–2026
- Mentors: Robin Pemantle, Yuliy Baryshnikov, Daniel E. Koditschek

University of Illinois Urbana-Champaign 2023–2024
- Postdoctoral Research Fellow at the Coordinated Science Laboratory
- Mentor: Yuliy Baryshnikov

Technion – Israel Institute of Technology 2020–2023
- Postdoctoral Research Fellow
- Mentors: Omer Bobrowski, Robert J. Adler

Education

New York University 2015–2020
- Ph.D. in Electrical Engineering
- Thesis title: Recovery of Permutational Data: Statistical Limits, Efficient Algorithms and Applications
- Advisor: Sundeep Rangan

Princeton University 2013–2015
- M.Sc. in Electrical Engineering
- Thesis title: Stochastic Block Model with Side Information
- Advisor: Emmanuel Abbe

Bilkent University 2008–2013
- B.Sc. in Electrical Engineering

Journal Publications & Preprints

Yuliy Baryshnikov and Efe Onaran. Ising disks: topology preserving glauber dynamics. *J. Appl. Comput. Topol.*, 10(3):Paper No. 16, 2026

Efe Onaran, Omer Bobrowski, and Robert J. Adler. Central limit theorems for local functionals of dynamic point processes. *Electronic Journal of Probability*, 31, 2026

Efe Onaran, Omer Bobrowski, and Robert J Adler. Functional central limit theorems for local statistics of spatial birth–death processes in the thermodynamic regime. *The Annals of Applied Probability*, 33(5):3958–3986, 2023

Efe Onaran and Soledad Villar. Shuffled linear regression through graduated convex relaxation. *arXiv:2209.15608*, 2022

Jinfeng Du, Efe Onaran, Dmitry Chizhik, Sivarama Venkatesan, and Reinaldo A Valenzuela. Gbps user rates using mmwave relayed backhaul with high-gain antennas. *IEEE Journal on Selected Areas in Communications*, 35:1363–1372, 2017

Efe Onaran, M Cengiz Onbasli, Alper Yesilyurt, Hyun Yong Yu, Ammar M Nayfeh, and Ali K Okyay. Silicon-germanium multi-quantum well photodetectors in the near infrared. *Optics express*, 20:7608–7615, 2012

In Preparation

Yuliy Baryshnikov and Efe Onaran. First-passage percolation under path constraints: efficiency–complexity tradeoffs for resource-limited agents.

In Conference Proceedings

Efe Onaran and Soledad Villar. Efficient belief propagation for graph matching. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing*, pages 9060–9064. IEEE, 2020

Efe Onaran and Soledad Villar. Projected power iteration for network alignment. In *Wavelets and Sparsity XVII*, page 103941C. International Society for Optics and Photonics, 2017

Efe Onaran, Siddharth Garg, and Elza Erkip. Optimal de-anonymization in random graphs with community structure. In *2016 50th Asilomar Conference on Signals, Systems and Computers*, pages 709–713. IEEE, 2016

Efe Onaran, Marios Gatzianas, and Christina Fragouli. Broadcast erasure channel with feedback: The two multicast case—algorithms and bounds. In *2013 International Symposium on Network Coding (NetCod)*, pages 1–6. IEEE, 2013

Talks

Joint Mathematics Meetings, Chicago	01/2027
Rutgers University, Applied and Computational Math Seminar	11/2026
AFOSR Dynamical Systems and Control Theory Review	08/2026
Penn/Temple Probability Seminar	10/2024
Bernoulli-IMS 11th World Congress in Probability and Statistics in Bochum	08/2024
CRM–PIMS Montreal Summer School in Probability – Short talk	07/2024
Koç University, Math Seminar – Online	04/2024
University of Illinois Urbana-Champaign, Probability Seminar	10/2023
Queen Mary University of London, Probability and Applications Seminar	12/2022
Random Structures and Algorithms in Gniezno, Poland	08/2022
Technion, Probability Seminar	04/2022
Purdue University, Probability Seminar	03/2022
International Conference on Acoustics, Speech, and Signal Processing – Online	05/2020
Asilomar Conference on Signals, Systems and Computers	09/2016
Sarnoff Symposium – Poster session	08/2016

Teaching

University of Pennsylvania, Department of Mathematics

- Instructor for Mathematics of Change - Part II (MATH 1080) — Fall 2025
- Co-instructor for Advanced Applied Probability (MATH 5460) — Spring 2025
- Co-instructor for Mathematics of Change - Part II (MATH 1080) — Fall 2024

University of Illinois Urbana-Champaign

- Instructor for Optimum Control Theory (ECE 553 - MATH 595) — Spring 2024

New York University

- Head TA for Applied Statistics (Undergraduate) — Spring 2020
- Head TA for Probability Theory (Undergraduate) — Fall 2019
- Head TA for Probability and Stochastic Processes (Graduate) — Spring 2016
- Head TA for Signal Processing (Undergraduate) — Fall 2015
- Grader for classes: Machine Learning, Optimization and Computational Linear Algebra, Inference and Representation, Mathematical Tools for Data Science

Service and Outreach

Reviewer for

- Annals of Applied Probability, Electronic Journal of Probability, Statistics and Probability Letters, Linear Algebra and its Applications, IEEE/ACM Transactions on Networking, Discrete Applied Mathematics, International Symposium on Information Theory

Tutoring

- High school math tutoring for Regents Exams, Opportunity Charter School, NY 2018

Honors and Awards

Weber Fellowship at NYU Tandon School of Engineering 2015-2016
Princeton University Department of Electrical Engineering Graduate Fellowship 2013-2014
Highest CGPA (3.97/4.00) among all 2013 graduates of Bilkent U., School of Eng. 2013
Full scholarship at Bilkent University 2008-2013
Prime Ministry of Turkey Scholarship 2008-2013

Citizenship

Türkiye, United States